

Asian Transport Database Overview

Summer 2026 Internship – Darcey Sutcliffe

Contents

1 Literature review

Completed a literature review to understand datasets/ complete work already published and open source.

Outputs:

Asia Transport Literature Tracker.xlsx

- Includes relevant publications, reports, datasets, organisations, list of countries in study area and modes relevant to each country.

Asia Transport Literature Tracker.pdf

- Summary of spreadsheet

2 Airports

2.1 Data sources used

2.2 Analysis done

Airport node spatial data correction (completed in Airport.py)

The World Bank airport dataset provided the only identified open source data containing flight connection paths between airports. Therefore, the World Bank airport points were used as the basis for the airport nodes used for the flight network.

However, the World Bank airport coordinates were found to have poor spatial accuracy. The airport locations were therefore checked and corrected using the OurAirports dataset, which appeared to provide better spatial accuracy.

The analysis used airport data from Ourairports.csv, worldbank_filtered_airport_flows.gpkg and worldbank_filtered_airport_volume.gpkg.

OurAirports data was first filtered to include only countries within the study area using Countries_list.xlsx. The World Bank airport coordinates were then corrected using the OurAirports coordinates where a matching IATA code was available. Where no matching OurAirports record was available, the original World Bank coordinates were retained.

The origin and destination coordinates of the World Bank flight connections were then corrected using the corrected airport coordinates.

airport_coordinate_audit.xlsx contains the audit of this matching process. The match_summary sheet provides a summary of the number of airport locations that were common between the two datasets. 14 airports in the World Bank dataset could not be matched and therefore could not be spatially corrected using OurAirports.

2.2.1 Shapefile matching (OSM_extract_country_loop and airport_shapefile_nearest.py):

Relevant tags were extracted from OSM.

For Aiports these were:

```
{"aeroway": ["aerodrome"]},
```

```
{"aeroway": ["terminal"]},
```

```
{"aeroway": ["runway"]},
```

```
{"aeroway": ["taxiway"]},
```

From here, any extracted OSM features located within 2 km of an airport node in `world-bank_filtered_airport_volume.gpkg` were assigned to that airport. The airport's IATA code was then added to the feature's attributes. The final output combines all extracted OSM airport features into a single layer, with attributes identifying the feature's original OSM tag (e.g. `aerodrome`, `terminal`, `runway` or `taxiway`) and the airport it was assigned to.

2.2.2 Still to do:

- Check the spatial location of the 14 airport nodes in `airport_flows_world_bank_corrected.gpkg`, which did not have a match to the OurAirports database. (Validating that the flows also update if any airports get updated)
- Add missing polygon data to the XX airports

3 Maritime

3.1 Data sources used/outputted

- Port watch
- Multi-hazard risk to global port infrastructure and resulting trade and logistics losses, [Jasper Verschuur](<https://www-nature-com.ezproxy-prd.bodleian.ox.ac.uk/articles/s43247-022-00656-7#auth-Jasper-Verschuur-Aff1>)
- Port_landuse.gpkg (shapefiles for ports)
- Global_maritime_network.gpkg (each node is at the centre of the port_landuse shapefiles that it is relevant to)
- Ports.shp

3.2 Analysis done

3.2.1 Maritime network (completed in `ports.py`):

Code initially copied from the `African-transport-dataset/scripts/preprocess/"ports_new_merge.py"`. Using the `Ports.shp` and updated 2023-2025 attribute data provided by Jasper Verschuur (`port_calls_average_2023-2025.csv`, `port_calls_average_2023-2025.csv`, `port_calls_average_2023-2025.csv`). A node and edge network was created, with the attributes. The network was broken up into 3 clusters, to aid visuals (`northern_russia`, `pacific_network` and `large_cluster`). The output contains two node types, ports (land) and maritime (networking), alongside the edge connections between.

-file output names

`global_maritime_network_PROVA_NEW1`

`asia & pacific_maritime_network_PROVA_NEW1`

`northern_russia_network_maritime_network`

`large_cluster_maritime_network`

`pacific_network_maritime_network`

3.2.2 Shapefile correction (all done using port_shapefile_nearest.py and manually):

port_landuse.gpkg shapefiles were matched to their corresponding ports by assigning the nearest port node to each polygon (nodes stored in asia_pacific_maritime_network_PROVA_NEW1.gpkg). The results of this automated matching are stored in port_landuse_nearest_port_id_lookup.xlsx.

The accuracy of these matches was then manually checked and recorded in port_landuse_nearest_port_id_lookup.xlsx. Each match was assessed using the distance between the port node and the land-use polygon, the names associated with both datasets, and the port's location on Google Maps. In the checked file, the checked column indicates whether the port land-use shapefile (Column A) has been matched to the correct port node (Column H). Where checked indicates that the match is incorrect, the correct port id column records the correct port ID, identified manually.

The corrections were then applied, and the port_landuse_nearest_port_ids.gpkg and port_landuse_nearest_port_ids.xlsx contain the correct results.

14 ports from the asia_pacific_maritime_network_PROVA_NEW1.gpkg file did not have any polygon data (from port_landuse.gpkg) matched to it. Seen in port_landuse_nearest_port_id_lookup_checked.xlsx sheet-node_polygon_audit, column missing_polygon_data = TRUE.

3.2.3 Still to do:

- Add missing polygon data to the 14 ports.

4 Road

5 Rail

6 Inland waterways

Investigated OSM tags, best appeared to be:

Tag = boat:yes

Tag = ferry : terminal

7 Multimodal

8 ARC

ARC is separated into two file directories:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/asia-transport-database
```

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/open-gira
```

9 Using open-gira on ARC

Purpose: Used to create the road and rail networks using OpenGIRA.

1. Go to the OpenGIRA directory:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/open-gira
```

2. Activate the OpenGIRA environment:

```

module load Mamba/23.11.0-0
export CONDA_ENVS_PATH=/home/mans4968/.mamba/envs
export MAMBA_ROOT_PREFIX=/home/mans4968/.mamba
eval "$(conda shell.bash hook)"
conda activate /home/mans4968/.mamba/envs/open-gira

```

3. Run OpenGIRA using Snakemake and the ARC profile (not run on a .sh file):

```
snakemake -profile config/ARC -cores 1
```

A specific output can also be requested, for example:

```
snakemake -profile config/ARC -cores 1 results/russia-latest_filter-road-primary/edges.
```

4. Check jobs currently running:

```
squeue -u $USER
```

5. Check completed/failed jobs:

```
sacct -u $USER -format=JobID,JobName,State,Elapsed,ExitCode
```

6. Outputs are stored in:

```
/data/cenv-opsis-astdb/{<USERNAME>}/open-gira/results/
```

Completed final networks are organised into:

```
results/road/
```

```
results/rail/
```

Other folders such as geoparquet/, slices/, json/ and input/ contain input or intermediate OpenGIRA data.

10 Asia-transport-database on ARC

Purpose: Used to run the Asia transport database processing, including the country-by-country OSM infrastructure extraction.

1. Go to the Asia database:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/asia-transport-database
```

2. Input OSM data are stored in:

```
incoming_data/osm/
```

3. Run the required Asia transport database script/job on ARC.

See various .sh files

4. Processed outputs are stored in:

```
processed_data/infrastructure/
```

11 Already existing country multimodal transport packages

Multimodal transport data packages have already been extracted for China, Singapore, New Zealand, and Australia. These datasets can be used later for comparison and validation against the wider transport database. The packages are stored under:

`incoming_data/infrastructure/[folder name]`

The content of each country package varies, and not all packages contain data for every transport mode.

12 Files Overview

Asia-transport-database repository

A catalogue of inputs, intermediate files, final outputs and processing scripts

Legend: entries marked with a blue star (★) are final outputs.

Files are grouped by their repository location; the descriptions below record their purpose, provenance and producing script.

1. \{\}*airport_flows_world_bank_corrected.gpkg Directory: `processed_data/infrastructure/airport/`

Description:

World Bank airport flow data with origin and destination coordinates corrected using the corrected airport locations.

Reference:

Derived from World Bank and OurAirports.

Produced by:

`Airports.py`

2. \{\}*airport_osm_features_within_2km.gpkg Directory: `processed_data/infrastructure/airport/`

Description:

OSM airport lines and polygons located within 2 km of corrected World Bank airport points and assigned to the relevant airport.

Reference:

Derived from OpenStreetMap and corrected World Bank airport data.

Produced by:

`airport_shapefile_nearest.py`

3. `airport_osm_features_within_2km.xlsx` Directory: `processed_data/infrastructure/airport/`

Description:

Audit workbook containing matched OSM features, airport summaries and OSM tag summaries.

Reference:

Derived from OpenStreetMap and corrected World Bank airport data.

Produced by:

airport_shapefile_nearest.py

4. airport_coordinate_audit.xlsx Directory: processed_data/infrastructure/airport/

Description:

Audit spreadsheet showing which World Bank airport locations could be matched to the OurAirports dataset and therefore spatially corrected. It also allows unmatched World Bank airport points to be identified.

Reference:

Derived from World Bank and OurAirports.

Produced by:

Airports.py

5. [country/region]_[tag].parquet Directory: processed_data/infrastructure/osm_filter/

Description:

OSM features extracted for specified tags from the country or regional PBF files. Outputs are produced for individual countries and, where specified, for the wider study area.

Reference:

Derived from Geofabrik OpenStreetMap extracts.

Produced by:

OSM_extract_country_loop.py

6. [country/region].osm.pbf Directory: incoming_data/osm/

Description:

Raw country or regional OpenStreetMap PBF extracts used as the source data for OSM feature extraction.

Reference:

<https://download.geofabrik.de/index.html>

Source/Produced by:

Geofabrik / OpenStreetMap

7. Countries_list.xlsx Directory: incoming_data/

Description:

List of all countries included in the study area.

Reference:

Created by Darcey Sutcliffe.

Source/Produced by:

8. filtered_Ourairports.csv Directory: processed_data/infrastructure/airport/

Description:

OurAirports data filtered to include small, medium and large airports within the study area.

Reference:

Derived from OurAirports.

Produced by:

Airports.py

9. filtered_Ourairports.gpkg Directory: processed_data/infrastructure/airport/

Description:

Spatial version of the filtered OurAirports data containing small, medium and large airports within the study area.

Reference:

Derived from OurAirports.

Produced by:

Airports.py

10. global_maritime_network_PROVA_NEW1.gpkg Directory: processed_data/infrastructure/port/

Description:

Global maritime network containing port and maritime network nodes and the connections between them.

Reference:

Derived from maritime port data and PortWatch attributes.

Produced by:

ports.py

11. \{*cleaned_asia_pacific_maritime_network_PROVA_NEW1.gpkg Directory: processed_data/infrastructure/port/

Description:

Final cleaned Asia-Pacific maritime network created by combining the three separate network clusters.

Reference:

Derived from the northern Russia, large cluster and Pacific maritime networks.

Produced by:

ports.py

12. asia_pacific_maritime_network_PROVA_NEW1.gpkg Directory: processed_data/infrastructure/port/

Description:

Asia-Pacific maritime network covering the study area and including connections between the separate network clusters.

Reference:

Derived from maritime port data and PortWatch attributes.

Produced by:

ports.py

13. large_cluster_maritime_network.gpkg Directory: processed_data/infrastructure/port/

Description:

Maritime network representing the large cluster containing the majority of the Asian network nodes.

Reference:

Derived from maritime port data and PortWatch attributes.

Produced by:

ports.py

14. northern_russia_network_maritime_network.gpkg Directory: processed_data/infrastructure/port/

Description:

Maritime network representing the northern Russia cluster.

Reference:

Derived from maritime port data and PortWatch attributes.

Produced by:

ports.py

15. pacific_network_maritime_network.gpkg Directory: processed_data/infrastructure/port/

Description:

Maritime network representing the Pacific cluster.

Reference:

Derived from maritime port data and PortWatch attributes.

Produced by:

ports.py

16. port_calls_average_2023-2025.csv Directory: incoming_data/Global port supply-chains/

Description:

Port attribute data containing average port calls for 2023–2025.

Reference:

PortWatch, provided by Jasper Verschuur.

<https://portwatch.imf.org/>

17. port_capacity_called_average_2023-2025.csv Directory: incoming_data/Global port supply-chains/

Description:

Port attribute data containing average port capacity called for 2023–2025.

Reference:

PortWatch, provided by Jasper Verschuur.

<https://portwatch.imf.org/>

18. port_landuse.gpkg Directory: incoming_data/infrastructure/port/

Description:

Polygon data representing maritime port land use.

Reference:

<https://data.mendeley.com/datasets/kdyt24tsh5/1>

Source:

Jasper Verschuur

19. port_landuse_nearest_port_id_lookup.xlsx Directory: processed_data/infrastructure/port/

Description:

Lookup showing the nearest port assigned to each port land-use polygon before manual corrections were applied.

Reference:

Derived from port land-use data and the maritime network.

Produced by:

port_shapefile_nearest.py

20. port_landuse_nearest_port_id_lookup_checked.xlsx Directory: incoming_data/infrastructure/port/

Description:

Manually checked spreadsheet assessing whether the nearest port assigned to each port land-use polygon was correct. Where an incorrect match was identified, the correct port_id was recorded.

Reference:

Created by Darcey Sutcliffe.

Source/Produced by:

Manual validation by Darcey Sutcliffe

21. port_landuse_nearest_port_id_lookup_checked_result.xlsx Directory: processed_data/infrastructure/port/

Description:

Validation spreadsheet showing the original and final port_id for each port land-use polygon, allowing confirmation that the manual corrections were applied correctly.

Reference:

Derived from the manually checked port lookup.

Produced by:

port_shapefile_nearest.py

22. \{*port_landuse_nearest_port_ids.gpkg Directory: processed_data/infrastructure/port/

Description:

Final GeoPackage containing the corrected relationship between each port land-use polygon and its corresponding port_id.

Reference:

Derived from port land-use data and the maritime network, with manual corrections applied.

Produced by:

port_shapefile_nearest.py

23. port_turn_around_time_average_2023-2025.csv Directory: incoming_data/Global port supply-chains/

Description:

Port attribute data containing average port turnaround time for 2023–2025.

Reference:

PortWatch, provided by Jasper Verschuur.

<https://portwatch.imf.org/>

24. Ports.csv Directory: incoming_data/infrastructure/port/

Description:

Point locations of maritime ports.

Reference:

<https://data.mendeley.com/datasets/kdyt24tsh5/1>

Source:

Jasper Verschuur

25. Ourairports.csv Directory: incoming_data/infrastructure/Airport/

Description:

OurAirports airport attributes and coordinates. Export of all global data reported by OurAirports.

Reference:

<https://ourairports.com/data/>

Source/Produced by:

External dataset

26. `utils_new.py` Directory: `scripts/preprocess/` (if applicable)

Description:

Utility functions used by the preprocessing scripts. The script was copied from the African-transport-dataset project.

Reference:

`African-transport-dataset/scripts/preprocess/`

Produced by:

Copied and adapted from the African-transport-dataset project.

27. `worldbank_filtered_airport_flows.gpkg` Directory: `incoming_data/infrastructure/Airport/`

Description:

World Bank airport flow data filtered to include only flows between airports within the study area.

Reference:

<https://datacatalog.worldbank.org/search/dataset/0038117/global-airports>

Source/Produced by:

External dataset

28. `worldbank_filtered_airport_volume.gpkg` Directory: `incoming_data/infrastructure/Airport/`

Description:

World Bank airport data filtered to include only countries within the study area.

Reference:

<https://datacatalog.worldbank.org/search/dataset/0038117/global-airports>

Source/Produced by:

External dataset

29. `\{*world_bank_airports_corrected.gpkg` Directory: `processed_data/infrastructure/airport/`

Description:

World Bank airport locations corrected using OurAirports coordinates where a matching IATA code was available. Original World Bank coordinates are retained where no matching OurAirports record was found.

Reference:

Derived from World Bank and OurAirports.

Produced by:
Airports.py